

Linux Basic Commands

1. File Operations

Introduction

File operations are the basic tasks performed on files in the Linux operating system. These commands allow users to create, view, copy, move, rename, search, and delete files. Every Linux user, programmer, system administrator, and cybersecurity professional uses these commands regularly. Learning file operations helps users manage files quickly and efficiently from the terminal.

1.1 ls Command

Purpose:

The `ls` command is used to display the contents of a directory.

Common Options:

Command	Description
<code>ls</code>	List files and folders
<code>ls -l</code>	Long listing format
<code>ls -a</code>	Show hidden files

ls -al	Show all files with details
ls -R	Show files recursively

Example:

```

(aishu@kali)-[~/Desktop]
$ ls
aishu

(aishu@kali)-[~/Desktop]
$ ls -l
total 4
drwxrwxr-x 2 aishu aishu 4096 Jul  3 11:28 aishu

(aishu@kali)-[~/Desktop]
$ ls -a
. .. aishu

(aishu@kali)-[~/Desktop]
$ ls -al
total 12
drwxr-xr-x  3 aishu aishu 4096 Jul  3 11:28 .
drwx----- 19 aishu aishu 4096 Jul  3 11:28 ..
drwxrwxr-x  2 aishu aishu 4096 Jul  3 11:28 aishu

(aishu@kali)-[~/Desktop]
$ la -R
.:
aishu

./aishu:

```

Real-Life Use:

Before opening or editing a file, users use **ls** to see available files.

1.2 pwd Command

Purpose

Shows the current working directory.

Example:

```
(aishu@kali)-[~/Desktop]
$ pwd
/home/aishu/Desktop
```

Real-Life Use:

It tells us exactly where you are in the Linux file system.

1.3 cd Command

Purpose

Changes the current directory.

Example:(Common Error)

```
(aishu@kali)-[~/Downloads]
$ cd ..

(aishu@kali)-[~]
$ cd Document
cd: no such file or directory: Document

(aishu@kali)-[~]
$ cd Documents
```

Real-Life Use:

The cd command is used to **move** from one folder to another.

1.4 touch Command

Purpose

Creates an empty file.

Example:

```
(aishu@kali)-[~/Documents]
└─$ touch note.txt

(aishu@kali)-[~/Documents]
└─$ ls
note.txt
```

Real-Life Use:

Creates a new empty file named **note.txt**.

1.5 cat Command

Purpose

Displays the content of a file.

Example:

```
(aishu@kali)-[~/Documents]
$ cat note.txt
hello aishu
```

Another Example:

Merge two files

```
(aishu@kali)-[~/Documents]
$ touch note1.txt

(aishu@kali)-[~/Documents]
$ touch note2.txt

(aishu@kali)-[~/Documents]
$ cat note.txt note1.txt > note2.txt

(aishu@kali)-[~/Documents]
$ ls
note1.txt note2.txt note.txt

(aishu@kali)-[~/Documents]
$ cat note2.txt
hello aishu
```

Real-Life Use:

Shows all text stored inside the file.

1.6 cp Command

Purpose

Copies files from one location to another.

Example:(Common Error)

```
(aishu@kali)-[~/Documents]
└─$ cp note.txt Downloads/
cp: cannot create regular file 'Downloads/': Not a directory

(aishu@kali)-[~/Documents]
└─$ cp note.txt Backup/
cp: cannot create regular file 'Backup/': Not a directory

(aishu@kali)-[~/Documents]
└─$ ls -l
total 8
-rw-rw-r-- 1 aishu aishu 0 Jul  3 12:00 note1.txt
-rw-rw-r-- 1 aishu aishu 12 Jul  3 12:01 note2.txt
-rw-rw-r-- 1 aishu aishu 12 Jul  3 11:58 note.txt

(aishu@kali)-[~/Documents]
└─$ mkdir backup

(aishu@kali)-[~/Documents]
└─$ cp note.txt backup/

(aishu@kali)-[~/Documents]
└─$ ls
backup note1.txt note2.txt note.txt

(aishu@kali)-[~/Documents]
└─$ cd backup

(aishu@kali)-[~/Documents/backup]
└─$ ls
note.txt
```

Real-Life Use:

Copies **note.txt** into the Backup folder.

1.7 mv Command

Purpose

Moves or renames files.

Example (Move):

```
(aishu@kali)-[~/Documents]
$ mkdir downloads

(aishu@kali)-[~/Documents]
$ mv note1.txt downloads

(aishu@kali)-[~/Documents]
$ ls
backup  downloads  note2.txt  note.txt

(aishu@kali)-[~/Documents]
$ cd downloads

(aishu@kali)-[~/Documents/downloads]
$ ls
note1.txt
```

Example (Rename):

```
(aishu@kali)-[~/Documents/downloads]
$ touch note5.txt

(aishu@kali)-[~/Documents/downloads]
$ mv note1.txt note5.txt

(aishu@kali)-[~/Documents/downloads]
$ ls
note5.txt
```

1.8 rm Command

Purpose:

Deletes files.

```
(aishu@kali)-[~/Documents/downloads]
$ rm note5.txt

(aishu@kali)-[~/Documents/downloads]
$ ls
```

1.9 find Command

Purpose

Searches for files or folders.

Example:

```
(aishu@kali)-[~/Documents]
$ find . -name note2.txt
./note2.txt

(aishu@kali)-[~/Documents]
$ cd ..

(aishu@kali)-[~]
$ find ~/Documents -name note2.txt
/home/aishu/Documents/note2.txt
```

1.10 file Command

Purpose

Shows the type of a file.

```
(aishu@kali)-[~]  
$ touch image.png  
  
(aishu@kali)-[~]  
$ file image.png  
image.png: empty
```

2. File Permissions

Introduction

Linux protects files using a permission system. Permissions decide who can read, write, or execute a file. This improves system security by preventing unauthorized access

```
(aishu@kali)-[~/Documents]  
$ ls -al  
total 24  
drwxr-xr-x  4 aishu aishu 4096 Jul  3 12:15 .  
drwx----- 19 aishu aishu 4096 Jul  3 12:30 ..  
drwxrwxr-x  2 aishu aishu 4096 Jul  3 12:09 backup  
drwxrwxr-x  2 aishu aishu 4096 Jul  3 12:20 downloads  
-rw-rw-r--  1 aishu aishu  12 Jul  3 12:01 note2.txt  
-rw-rw-r--  1 aishu aishu  12 Jul  3 11:58 note.txt
```

Permission Types

Permission	Meaning
r	Read
w	Write
x	Execute

Permission Values

Permission	Value
Read (r)	4
Write (w)	2
Execute (x)	1

```
(aishu@kali)-[~/Documents]
$ ls -al
total 24
drwxr-xr-x  4 aishu aishu 4096 Jul  3 12:15 .
drwx----- 19 aishu aishu 4096 Jul  3 12:30 ..
drwxrwxr-x  2 aishu aishu 4096 Jul  3 12:09 backup
drwxrwxr-x  2 aishu aishu 4096 Jul  3 12:20 downloads
-rw-rw-r--  1 aishu aishu  12 Jul  3 12:01 note2.txt
-rw-rw-r--  1 aishu aishu  12 Jul  3 11:58 note.txt
```

```
(aishu@kali)-[~/Documents]
$ chmod 755 note.txt
```

```
(aishu@kali)-[~/Documents]
$ ls -al
total 24
drwxr-xr-x  4 aishu aishu 4096 Jul  3 12:15 .
drwx----- 19 aishu aishu 4096 Jul  3 12:30 ..
drwxrwxr-x  2 aishu aishu 4096 Jul  3 12:09 backup
```

```
drwxrwxr-x 2 aishu aishu 4096 Jul 3 12:20 downloads
-rw-rw-r-- 1 aishu aishu 12 Jul 3 12:01 note2.txt
-rwxr-xr-x 1 aishu aishu 12 Jul 3 11:58 note.txt
```

Before chmod 755 note.txt:

User	Permissions
Owner	rw- (Read, Write)
Group	rw- (Read, Write)
Others	r-- (Read)

After chmod 755 note.txt

User	Permissions
Owner	rwx (Read, Write, Execute)
Group	r-x (Read, Execute)
Others	r-x (Read, Execute)

Difference

Before	After	Change
Owner: rw-	rwx	Execute (x) permission added
Group: rw-	r-x	Write (w) removed, Execute (x) added

Others: r--	r-x	Execute (x) permission added
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KEY DIFFERENCE IS:

After The owner can read, write, and execute the file, while the group and others can read and execute it but cannot modify it.